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EXPERIMENT: COEFFICIENT OF FRICTION

Objectives:

1. To determine the coefficient of static (μ_s) and kinetic (μ_k) friction.
2. To understand the differences between the characteristics of static friction and kinetic friction.

EXPERIMENT SUMMARY

INTRODUCTION

Friction is one of the most common forces encountered in everyday life and various mechanical systems. It arises when two surfaces are in contact and tend to move relative to one another. According to Halliday, Resnick, and Walker [1], friction is a contact force that always acts opposite to the direction of motion or the tendency of motion of an object. The magnitude of the frictional force depends on the characteristics of the contacting surfaces and the normal force acting between them.

In general, friction is classified into two types static friction and kinetic friction. Static friction acts when an object is at rest and prevents it from moving until it reaches its maximum value. In contrast, kinetic friction acts when an object moves relative to the contacting surface. The coefficient of static friction is generally greater than the coefficient of kinetic friction because a greater force is required to initiate motion than to maintain the motion of an object that is already moving [2].

The coefficient of friction can be determined using either the static method or the kinetic method. In the static method, the coefficient of static friction is determined under the condition that the object is just about to move due to the applied pulling force. For a system on a horizontal surface, where the object's mass is m_1 and the hanging load's mass is m_2 , the coefficient of static friction is expressed as:

$$\mu_s = \frac{m_2}{m_1} \quad (1)$$

Where μ_s is the coefficient of static friction, m_1 is the mass of the object on the surface (kg) and m_2 is the mass of the hanging load (kg) as shown in Figure 1. If the experiment is performed on an inclined plane with an inclination angle of θ , the force analysis based on Newton's Second Law yields the following relationship:

$$\mu_s = \frac{m_2}{m_1} \sec \theta - \tan \theta \quad (2)$$

Where θ is the angle of inclination of the plane with respect to the horizontal. In the kinetic method, the coefficient of kinetic friction is determined when the object is in motion, causing the system to have a certain acceleration. For a horizontal surface, the relationship between the coefficient of kinetic friction, the masses of the system, and the acceleration is expressed as:

$$\mu_k = \frac{m_2}{m_1} - \frac{(m_1 + m_2)a}{m_1 g} \quad (3)$$

Where μ_k is the coefficient of kinetic friction, m_1 is the mass of the object on the inclined plane (kg), m_2 is the mass of the hanging load (kg), a is the acceleration of the system (m/s^2) and g is the acceleration due to gravity. For an inclined plane, by considering the components of the gravitational force parallel and perpendicular to the plane, the following equation is obtained:

$$\mu_k = \frac{m_2}{m_1} \left(\sec \theta - \frac{a}{g \cos \theta} \right) - \left(\tan \theta + \frac{a}{g \cos \theta} \right) \quad (4)$$

Where θ is the angle of inclination of the plane with respect to the horizontal. This equation shows that the coefficient of kinetic friction depends not only on the masses of the objects but also on the system's acceleration and the angle of inclination of the plane.

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 1. Tools and materials for the coefficient of friction experiment

Apparatus	Quantity	Function
Friction board	1 set	Used as the track on which the wooden block moves during the experiment. It can also be inclined for measurements at specific angles.
Wooden block	1 piece	Used as the test object subjected to friction while moving on the friction board.
Loads	1 set	Used to provide different mass variations that generate the pulling force on the wooden block through the hanging pan.
Pan	1 piece	Used to hold the hanging weights.
Digital scale	1 piece	Used to measure the masses of the wooden block, the weights, and the hanging pan.
Stopwach	1 piece	Used to measure the time required for the wooden block to travel a distance of 1 m.
Spirit level	1 piece	Used to ensure that the friction board is perfectly horizontal during the horizontal surface experiment.
Protractor	1 piece	Used to measure and adjust the inclination angle of the friction board to 10° during the inclined plane experiment.

B. Procedure

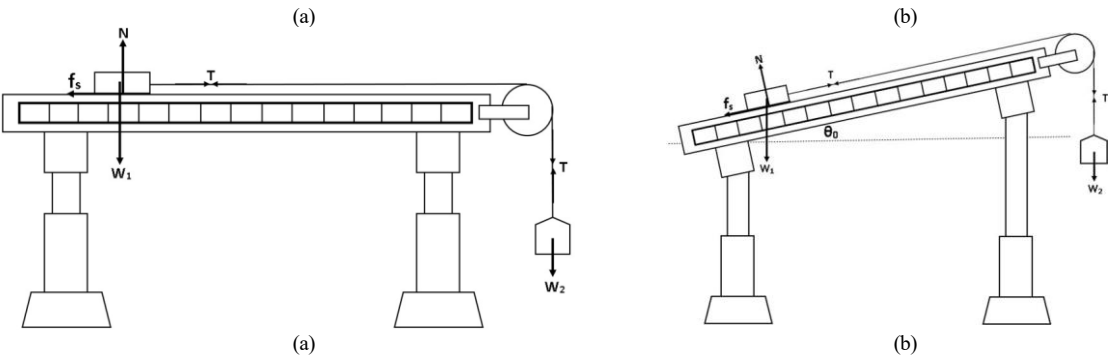


Figure 1. (a) Experimental setup for the coefficient of static and kinetic friction at an angle of 0° and (b) at an angle of θ_0

This experiment is conducted with the following step.

A. Static Method

1. Arrange the equipment according to the schematic in Figure 1a. Place the wooden block on the friction board over 1 meter.
2. Add loads to the pan until the system is on the verge of moving.
3. Measure the mass of the wooden block and the mass of the loads (including the pan) using a scale, only one data point is needed for the wooden block’s mass.
4. Repeat the previous steps with an inclination angle of $\theta = 10^\circ$ as shown in the schematic in Figure 1b.
5. Perform the experiment again with 5 repetitions using different combinations of load masses.

B. Kinetic Method

1. Place the wooden block on the track over 1 meter, then add a load to the pan until the system moves with an acceleration, a . Use the same mass on the block and in the pan for each repetition.
2. Record the time it takes for the system to travel that distance, then repeat the experiment five

times.

3. Repeat the steps above using an inclination angle of $\theta_0 = 10^\circ$.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement results obtained in the laboratory:

Table 2. Data results for the measurement of μ_s over a 1 m distance

Repetition	Horizontal surface		Inclined plane		
	m_1 (kg)	m_2 (kg)	angle	m_1 (kg)	m_2 (kg)
1					
2					
3					
4					
5					
Average					

Table 3. Data results for the measurement of μ_k over a 1 m distance

Repetition	Horizontal surface			Inclined plane			
	m_1 (kg)	m_2 (kg)	t (s)	angle	m_1 (kg)	m_2 (kg)	t (s)
1							
2							
3							
4							
5							
Average							

Approved by:

Laboratory Assistant

B. Calculation Analysis

(Calculate the coefficients of static and kinetic friction using Equations 1, 2, 3, and 4!)

The following is a sample calculation based on experimental data.

Sample Calculation:

The final calculation results for all data obtained are as follows.
 Table 4. Calculations results for the measurement of μ_s over a 1 m distance

Repetition	Horizontal surface	Inclined plane
1		
2		
3		
4		
5		
Average		

Table 5. Calculations results for the measurement of μ_k over a 1 m distance

Repetition	Horizontal surface			Inclined plane		
	t (s)	a (m/s^2)	μ_k	t (s)	a (m/s^2)	μ_k
1						
2						
3						
4						
5						
Average						

C. Discussions

CONCLUSIONS

From this experiment, it can be concluded that:

1.

2.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, Fundamentals of Physics Extended, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [2] R. A. Serway and J. W. Jewett, Physics for Scientists and Engineers with Modern Physics, 9th ed. Mason, OH, USA: Brooks/Cole, Cengage Learning, 2012.

PRELIMINARY ASSIGNMENT

1. Explain the definitions of static friction and kinetic friction, and describe the differences between them.
2. Explain the factors that affect the magnitude of the frictional force acting on an object.
3. Why is the coefficient of static friction (μ_s) generally greater than the coefficient of kinetic friction (μ_k)?
4. How is Newton's Second Law applied to determine the coefficient of friction using the static and kinetic methods?